Matrix-free reduction of the HE-regression normal equations

Notation and standing assumptions

- N individuals, C cell types, p fixed-effect columns; $X \in \mathbb{R}^{(NC)\times p}$ has full column rank p.
- Row indexing is *individual-major*: row $\overline{nc} := nC + c$ corresponds to individual n, cell type c. Write $X_{:,c,:}$ for the $N \times p$ submatrix of rows with cell-type index c.
- $A_{\text{inv}} := (X^\top X)^{-1}, \quad H := X A_{\text{inv}} X^\top, \quad \text{proj} := I_{NC} - H.$
- For symmetric $M \in \mathbb{R}^{N\times N}$ and symmetric $F \in \mathbb{R}^{C\times C}$: $Q(M,F) := \text{proj}\,\text{kron}(M,F)\,\text{proj}$.
- Frobenius inner product: $\langle U, V\rangle_F := \text{tr}(U^\top V)$.

Lemma 1 (structure identity)

For any $M \in \mathbb{R}^{N\times N}$, $F \in \mathbb{R}^{C\times C}$:

$$X^\top \text{kron}(M,F)\, X \;=\; \sum_{c,d} F[c,d]\cdot \left(X_{:,c,:}^\top\, M\, X_{:,d,:}\right) \;=:\; \text{Pmat}_M(F). \tag{1}$$

Proof. By definition of the Kronecker product under individual-major indexing, $\text{kron}(M,F)[\overline{nc},\overline{md}] = M[n,m]\,F[c,d]$. Expanding the matrix product entrywise, for $q, r \in \{1,\dots,p\}$:

$$(\text{kron}(M,F)X)[\overline{nc}, r] = \sum_{m,d} M[n,m]\,F[c,d]\,X[\overline{md}, r],$$

so

$$(X^\top \text{kron}(M,F)X)[q,r] = \sum_{n,c} X[\overline{nc},q] \sum_{m,d} M[n,m]F[c,d]X[\overline{md},r] = \sum_{c,d} F[c,d] \sum_{n,m} X[\overline{nc},q]\,M[n,m]\,X[\overline{md},r],$$

and the innermost double sum is exactly the (q,r) entry of $X_{:,c,:}^\top M X_{:,d,:}$. This holds for every q, r, proving (1) as a matrix identity. ∎

Theorem 1 (QQ reduction)

For symmetric $A, B \in \mathbb{R}^{N\times N}$ and symmetric $F, G \in \mathbb{R}^{C\times C}$:

$$\begin{aligned}
\langle Q(A,F), Q(B,G)\rangle_F \;&=\; \operatorname{tr}(\operatorname{proj}\operatorname{kron}(A,F)\operatorname{proj}\operatorname{kron}(B,G)) \\
&= \operatorname{tr}\big((I-H)\operatorname{kron}(A,F)\,(I-H)\operatorname{kron}(B,G)\big) \\
&= \operatorname{tr}\big((\operatorname{kron}(A,F)-H\operatorname{kron}(A,F))\,(\operatorname{kron}(B,G)-H\operatorname{kron}(B,G))\big) \\
&= \operatorname{tr}(\operatorname{kron}(A,F)\operatorname{kron}(B,G)) - 2\,\operatorname{tr}(H\operatorname{kron}(A,F)\operatorname{kron}(B,G)) \\
&\qquad + \operatorname{tr}(H\operatorname{kron}(A,F)\,H\operatorname{kron}(B,G)) \\
&= \operatorname{tr}(AB)\,\operatorname{tr}(FG) \\
&- 2\operatorname{tr}\big(A_{\text{inv}}\operatorname{Pmat}_{AB}(FG)\big) \\
&+ \operatorname{tr}\big(A_{\text{inv}}\operatorname{Pmat}_A(F)\,A_{\text{inv}}\operatorname{Pmat}_B(G)\big). \qquad (2)
\end{aligned}$$

Theorem 2 (target-term reduction)

Let $D = \operatorname{diag}(\operatorname{vec}(\text{ctnu}))$, $y_p = \operatorname{proj} y$, Y_p its $N \times C$ reshape, and

$$t \;:=\; y_p y_p^\top - \operatorname{proj} D \operatorname{proj}.$$

Then $\langle Q(A,F), t\rangle_F = \text{(i)} - \text{(ii)}$, where:

(i)

$$\langle Q(A,F), y_p y_p^\top\rangle_F = \sum_{c,d} F[c,d]\cdot\big(Y_{p,:,c}^\top A\, Y_{p,:,d}\big). \qquad (3)$$

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(ii)

$$\begin{aligned}
\langle Q(A,F), \operatorname{proj} D \operatorname{proj}\rangle_F &= \operatorname{tr}(Q(A,F)\operatorname{proj} D \operatorname{proj}) \\
&= \operatorname{tr}(M\operatorname{proj} D \operatorname{proj}), \qquad M := \operatorname{kron}(A,F), \\
&= \operatorname{tr}(M(I-H)D(I-H)) \\
&= \operatorname{tr}(DM) - 2\,\operatorname{tr}(HDM) + \operatorname{tr}(HDHM) \\
&= \operatorname{tr}(DM) - 2\operatorname{tr}\big(A_{\text{inv}}\,[X^\top DMX]\big) + \operatorname{tr}(A_{\text{inv}}\,X^\top DX A_{\text{inv}} X^\top MX). \qquad (4)
\end{aligned}$$

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